UNIVERSITI MALAYSIA TERENGGANU

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

CSE3023
WEB-BASED APPLICATION DEVELOPMENT

LAB REPORT 2

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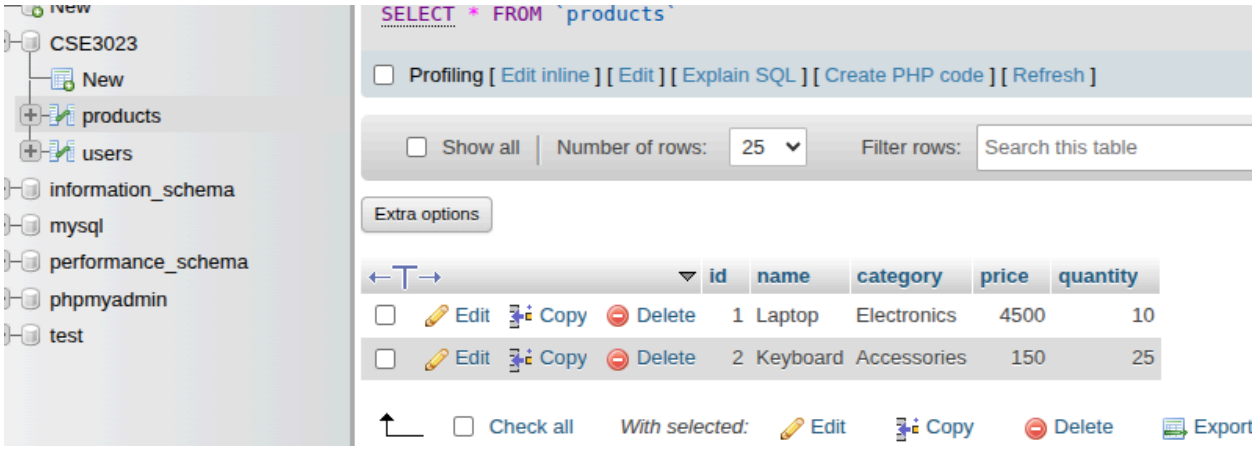
2.9 Exercise

Database Setup:

```sql
USE CSE3023;
CREATE TABLE IF NOT EXISTS products (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    category VARCHAR(50),
    price DOUBLE NOT NULL,
    quantity INT NOT NULL
);

INSERT INTO products (name, category, price, quantity) VALUES
('Laptop', 'Electronics', 4500.00, 10),
('Keyboard', 'Accessories', 150.00, 25);
```



JavaBean (Product.java):

```java
package com.lab.model;

public class Product {
    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    public Product() {}

    public Product(int id, String name, String category, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    public Product(String name, String category, double price, int quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    // Getters and Setters
    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getCategory() { return category; }
    public void setCategory(String category) { this.category = category; }
    public double getPrice() { return price; }
    public void setPrice(double price) { this.price = price; }
    public int getQuantity() { return quantity; }
    public void setQuantity(int quantity) { this.quantity = quantity; }
}
```

Data Access Object (DAO): ProductDAO.java

index.html × | Product.java × | ProductDAO.java × | add_product.html × | ViewProductServlet.java × | InsertProductServlet.java × | DeleteProductServlet.java × | UpdateP

Source | History

```java
package com.lab.dao;

import com.lab.model.Product;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class ProductDAO {
    private String jdbcURL = "jdbc:mysql://localhost:3307/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";

    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);
        } catch (SQLException | ClassNotFoundException e) { e.printStackTrace(); }
        return connection;
    }

    public void insertProduct(Product product) throws SQLException {
        String sql = "INSERT INTO products (name, category, price, quantity) VALUES (?, ?, ?, ?)";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.executeUpdate();
        }
    }

    public List<Product> selectAllProducts() {
        List<Product> products = new ArrayList<>();
        String sql = "SELECT * FROM products";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            ResultSet rs = pstmt.executeQuery();
            while (rs.next()) {
                products.add(new Product(rs.getInt("id"), rs.getString("name"),
                             rs.getString("category"), rs.getDouble("price"), rs.getInt("quantity")));
            }
        } catch (SQLException e) { e.printStackTrace(); }
        return products;
    }

    public boolean deleteProduct(int id) throws SQLException {
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement("DELETE FROM products WHERE id = ?")) {
            pstmt.setInt(1, id);
```

```java
            pstmt.setInt(1, id);
            return pstmt.executeUpdate() > 0;
        }
    }

    public Product selectProduct(int id) {
        Product product = null;
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement("SELECT * FROM products WHERE id = ?")) {
            pstmt.setInt(1, id);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                product = new Product(id, rs.getString("name"), rs.getString("category"), rs.getDouble("price"), rs.getInt("quantity"));
            }
        } catch (SQLException e) { e.printStackTrace(); }
        return product;
    }

    public boolean updateProduct(Product product) throws SQLException {
        String sql = "UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.setInt(5, product.getId());
            return pstmt.executeUpdate() > 0;
        }
    }
}
```

InsertProductDAO.java:

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/InsertProductServlet")
public class InsertProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    @Override
    public void init() {
        productDAO = new ProductDAO();
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product newProduct = new Product(name, category, price, quantity);
        try {
            productDAO.insertProduct(newProduct);
            response.sendRedirect("ViewProductServlet");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

ViewProductDAO:

```java
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/ViewProductServlet")
public class ViewProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    @Override
    public void init() {
        productDAO = new ProductDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        List<Product> list = productDAO.selectAllProducts();

        out.println("<html><body>");
        out.println("<h2>Product Inventory System</h2>");
        out.println("<table border='1' cellpadding='5'>");
        out.println("<tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Qty</th><th>Actions</th></tr>");

        for (Product p : list) {
            out.println("<tr>");
            out.println("<td>" + p.getId() + "</td>");
            out.println("<td>" + p.getName() + "</td>");
            out.println("<td>" + p.getCategory() + "</td>");
            out.println("<td>" + String.format("%.2f", p.getPrice()) + "</td>");
            out.println("<td>" + p.getQuantity() + "</td>");
            out.println("<td>");
            out.println("<a href='UpdateProductServlet?id=" + p.getId() + "'>Edit</a> | ");
            out.println("<a href='DeleteProductServlet?id=" + p.getId() + "' onclick='return confirm(\"Delete this product?\")'>Delete</a>");
            out.println("</td>");
            out.println("</tr>");
        }

        out.println("</table>");
        out.println("<br><a href='add_product.html'>Add New Product</a>");
        out.println("</body></html>");
    }
```

DeleteProductDAO:

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/DeleteProductServlet")
public class DeleteProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    @Override
    public void init() {
        productDAO = new ProductDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        int id = Integer.parseInt(request.getParameter("id"));
        try {
            productDAO.deleteProduct(id);
            response.sendRedirect("ViewProductServlet");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

UpdateProductServlet.java:

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/UpdateProductServlet")
public class UpdateProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    @Override
    public void init() {
        productDAO = new ProductDAO();
    }

    // 1. Show existing data in a form
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        int id = Integer.parseInt(request.getParameter("id"));
        Product p = productDAO.selectProduct(id);

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html><body><h2>Update Product</h2>");
        out.println("<form action='UpdateProductServlet' method='POST'>");
        out.println("<input type='hidden' name='id' value='" + p.getId() + "'>");
        out.println("Name: <input type='text' name='name' value='" + p.getName() + "' required><br><br>");
        out.println("Category: <input type='text' name='category' value='" + p.getCategory() + "'><br><br>");
        out.println("Price: <input type='number' step='0.01' name='price' value='" + p.getPrice() + "' required><br><br>");
        out.println("Quantity: <input type='number' name='quantity' value='" + p.getQuantity() + "' required><br><br>");
        out.println("<input type='submit' value='Save Changes'>");
        out.println("</form></body></html>");
    }
```

```java
    // 2. Process the update
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product updatedProduct = new Product(id, name, category, price, quantity);
        try {
            productDAO.updateProduct(updatedProduct);
            response.sendRedirect("ViewProductServlet");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

User Interface (View): add_product.html

```html
<!DOCTYPE html>
<html>
<body>
    <h2>Add New Product</h2>
    <form action="InsertProductServlet" method="POST">
        Name: <input type="text" name="name" required><br><br>
        Category: <input type="text" name="category"><br><br>
        Price: <input type="number" step="0.01" name="price" required><br><br>
        Quantity: <input type="number" name="quantity" required><br><br>
        <input type="submit" value="Add Product">
    </form>
</body>
</html>
```

OUTPUT:

Add New Product

Name: SSD

Category: Electronics

Price: 1700

Quantity: 20

Add Product

Product Inventory System

ID	Name	Category	Price	Qty	Actions
1	Laptop	Electronics	4500.00	10	Edit | Delete
2	Keyboard	Accessories	150.00	250	Edit | Delete
4	RAM	Electronics	1000.00	2	Edit | Delete
5	SSD	Electronics	1700.00	20	Edit | Delete

Add New Product

←T→			id	name	category	price	quantity
Edit	Copy	Delete	1	Laptop	Electronics	4500	10
Edit	Copy	Delete	2	Keyboard	Accessories	150	250
Edit	Copy	Delete	4	RAM	Electronics	1000	2
Edit	Copy	Delete	5	SSD	Electronics	1700	20

Check all *With selected:* Edit Copy Delete E

Update:

Update Product

Name: SSD-2

Category: Electronics

Price: 1700.0

Quantity: 20

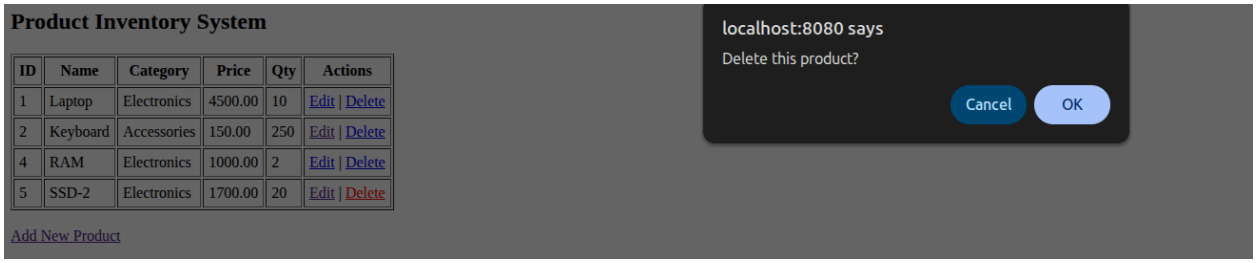
Save Changes

Product Inventory System

ID	Name	Category	Price	Qty	Actions
1	Laptop	Electronics	4500.00	10	Edit | Delete
2	Keyboard	Accessories	150.00	250	Edit | Delete
4	RAM	Electronics	1000.00	2	Edit | Delete
5	SSD-2	Electronics	1700.00	20	Edit | Delete

Add New Product

Delete:



Product Inventory System

ID	Name	Category	Price	Qty	Actions
1	Laptop	Electronics	4500.00	10	Edit | Delete
2	Keyboard	Accessories	150.00	250	Edit | Delete
4	RAM	Electronics	1000.00	2	Edit | Delete

Add New Product